

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SAVEETHA COMPILER ASSESSMENT (LEVEL 1 & VIVA VOCE)

Course Code:	CSA09	Course Title:	Programming in Java
CO Assessed:	CO1 – Manipulation (BL4)	Assessment:	AT1 – Saveetha Compiler (Level 1 & Viva)
Weightage:	20% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	CO1: <i>Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)</i>		
Student Name:	N. Hemanth Reddy	Reg. No.:	192472215
Section / Batch:	SLOT B / 2024	Date:	27/07/2026

Instructions to Students

Answer ALL 5 questions. Total: 100 Marks.

Questions 1–5: Write complete, syntactically correct Java programs on the Saveetha Compiler with proper indentation. Each question carries 20 Marks.

For program questions: test with at least two sample inputs; ensure output is displayed clearly. Each question carries 5 Marks.

No calculators or electronic devices permitted during the viva session.

Question Type	Description	Questions	Marks
Level 1 – Program Writing	Control Flow, classes and Object programs, Collection Framework on compiler	Q1 – Q4	50
GRAND TOTAL:			100 Marks

SECTION A – LEVEL 1: PROGRAM WRITING QUESTIONS (Q1–Q5)

Q1. (Level 1 – Program Writing) [20 Marks]

START

Read Principal
Read Years
Read Choice

IF Choice = 'y' OR Choice = 'Y'
Date = 12

Answer / Program with output:

Secure Exit

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level 1 Exam
 Level Exam `</> Java` 24 Jul 2026, 02:50 PM

QUESTIONS

LEVEL 1 — EASY

Q1 Numbers

Q2 Loops

Q3 Strings

Q4 Array

Q5 Control

5 / 5 submitted

Level 1 — Easy (E) 51 — Control

Submitted

Write a program using function to calculate the simple interest. Suppose the customer is a senior citizen. He is being offered 12 percent rate of interest; for all other customers, the ROI is 10 percent.

Sample Input:

Enter the principal amount: 200000

Enter the no of years: 3

Is customer senior citizen (y/n): n

Sample Output:

Interest: 60000

1. Principal: 2000 , Years: 0
 2. Principal: 20000 , Years: -2
 3. Principal: -2000 , Years: 2
 4. Principal: 2 , Years: 2000
 5. Principal: 0 , Years: 5

Your Code

```

1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double p = sc.nextDouble();
6         int y = sc.nextInt();
7         char c = sc.next().charAt(0);
8         double r;
9         if (c == 'y' || c == 'Y')
10            r = 12;
11         else
12            r = 10;

```

Output

```

Interest: 60000

```

Q2. (Level 1 – Program Writing) [20 Marks]

START

Declare Matrix A

Declare Matrix B

FOR i = 0 to 1

Answer / Program with output:

Secure Exit

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QUESTIONS

LEVEL 1 — EASY

Q1 Numbers

Q2 Loops

Q3 Strings

Q4 Array

Q5 Control

5 / 5 submitted

Level 1 — Easy (E) Matrix Addition — Array

Submitted

Write a program for matrix addition?

Sample Input:

Mat1 =

1 2

5 3

Mat2 =

2 3

4 1

Sample Output:

Mat Sum =

3 5

9 4

Your Code

```

1 class main {
2     public static void main(String[] args) {
3         int a[][] = {{1,2},{5,3}};
4         int b[][] = {{2,3},{4,1}};
5         System.out.println("Mat Sum =");
6         for (int i=0;i<2;i++){
7             for(int j=0;j<2;j++){
8                 System.out.print((a[i][j]+b[i][j])+" ");
9                 System.out.println();
10            }
11        }
12    }

```

Output

```

Mat Sum =
3 5
9 4

```

Q3. (Level 1 – Program Writing) [20 Marks]

START

Read Word

Print "Vowels"

FOR each character in Word

IF character is a,e,i,o,u

Print character

Answer / Program with output:



Secure Exit

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QUESTIONS

LEVEL 1 — EASY

Q1
Numbers

Q2
Loops

Q3
Strings

Q4
Array

Q5
Control

5 / 5 submitted

Level 1 — Easy (E) Vow's-Con's — Strings

Submitted

Write a program to print consonants and vowels separately in the given word
Sample Input:
Given Word: Engineering
Sample Output:
Consonants: n g n r n g
Vowels: e i e e i

1. TRY
2. MEDIAN
3. ONE
4. KNOWLEDGE
5. EDUCATION

Your Code

```
1 class Main {
2     public static void main(String[] args) {
3         String s = "Engineering";
4         for(int i=0;i<s.length();i++) {
5             char c=s.charAt(i);
6             if(c=='a' || c=='e' || c=='o' || c=='u' ||
7               c=='A' || c=='E' || c=='O' || c=='U')
8                 System.out.print(c+" ");
9         }
10    }
11 }
```

Output

```
E e e
```

Q4. (Level 1 – Program Writing) [20 Marks]

START

Read Character
Read Rows

FOR i = 1 to Rows
FOR j = 1 to i
Print Character

Answer / Program with output:



Secure Exit

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QUESTIONS

LEVEL 1 — EASY

Q1
Numbers

Q2
Loops

Q3
Strings

Q4
Array

Q5
Control

5 / 5 submitted

Level 1 — Easy (E) RT-Pattern — Loops

Submitted

Write a program to print the following pattern
Sample Input:
Enter the Character to be printed: %
Max Number of time printed: 3
%
% %
% % %

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         char ch = sc.next().charAt(0);
6         int n = sc.nextInt();
7         for (int i=1; i<=n; i++) {
8             for (int j=1; j<=i; j++)
9                 System.out.print(ch + " ");
10            System.out.println();
11        }
12    }
13 }
```

Output

```
%
% %
% % %
```

Previous

Next

Q5. (Level 1 – Program Writing) [20 Marks]

START

Read Number

Root = SquareRoot(Number)

Print Root

Print -Root

Answer / Program with output:

Secure Exit**SIMATS Online Programming Lab**JAVA PROGRAMMINGNAGIREDDY HEMANTH REDDY
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QUESTIONS
LEVEL 1 — EASY
Q1 Numbers ✓
Q2 Loops ✓
Q3 Strings ✓
Q4 Array ✓
Q5 Control ✓
5 / 5 submitted

Level 1 — Easy (E) Sqrt — Numbers ✓ Submitted
Write a program to find the square root of a perfect square number(print both the positive and negative values)
Sample Input:
Enter the number : 6561
Sample Output:
Square Root: 81, -81
1. 1225
2. 9801
3. 1827
4. -100
5. 0
Your Code

```
1 import java.util.Scanner;  
2 public class main {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         int n = sc.nextInt();  
6         int root = (int)Math.sqrt(n);  
7         System.out.println(root + ", -" + root);  
8     }  
9 }
```

Output

```
81, -81
```

RUBRICS FOR EVALUATION

Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (3–4 Marks)	Level 2 – Developing (2 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Analysis & Understanding				
Logic / Algorithm Design				
Python Syntax & String Operations				
Output Correctness & Testing				
Communication & Explanation (Viva)				

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____